Yuliang Peng

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EDUCATION

University of Maryland, College Park
- Bachelor of Science in Computer Science (Dean's List) **Graduated : May 2024**
- Master of Science in Machine Learning **Expected Graduation: May 2027**
- Relevant Courses: Machine Learning, Computer Vision, Robotics, Deep Learning, Data Structures and Algorithms

WORK EXPERIENCE / RESEARCH EXPERIENCE

University of Maryland, College Park *Research Assistant* **Maryland, USA** Nov 2025 - Present

- Developed GPU-accelerated spatiotemporal embeddings for high-rate event camera streams, enabling efficient ML inference with optimized CUDA-based parallelization.
- Conducted data-driven analysis of aviation network resilience using network-level modeling on Delta system outage data.
- Applied NLP and sentiment analysis to TripAdvisor low-altitude travel reviews to extract passenger perception dimensions

ModeSens *Software Engineer Full-Time* **Remote, USA** Jun 2025 - Sep 2025

- Built a distributed e-commerce crawling system using asynchronous pipelines and serverless triggers, improving product coverage by 15%.
- Developed an NLP-based search-term normalization and translation pipeline to align user queries with merchant catalogs, improving search precision by 23%.
- Implemented an image-based product matching workflow that compares user-uploaded images against internal product catalogs to support visual search and recommendations.

Bank of China, USA *Software Engineer Full-Time* **Los Angeles, USA** July 2024 - May 2025

- Led bug bash and reliability efforts for internal enterprise systems, resolving 70+ high-impact issues and reducing productivity-blocking tickets by 30%.
- Implemented a secure client authorization framework using cryptographic signatures and the Chain of Responsibility pattern, adopted by 15+ internal teams.
- Built AWS-based data pipelines (EC2, S3, RDS) for real-time ingestion and storage of financial transaction data, improving data reliability and operational efficiency.

Air China *Software Engineer Intern* **Sichuan,China** May 2023 - August 2023

- Developed Unix shell scripts to automate system backups across Linux environments, reducing manual workload by 40%.
- Revamped the secure login system with mobile QR code scanning, streamlining user identification. (adopted in 2024)

PROJECTS

Tabit - AI-Powered Chrome Extension | Google Chrome Extension Hackathon | Github Code

- Developed a Chrome extension using React, Vite, and Manifest V3 to support tab batching, session persistence, and history-based tab organization.
- Implemented tab grouping based on similarity scoring and hierarchical clustering, with Web Worker–based parallel processing to handle 100+ open tabs.
- Built a lightweight memory mechanism to summarize and reuse users' historical tab sessions, along with a reminder feature to surface previously saved contexts.

Image-Based Product Matching Tool | Machine Learning | Github Code

- Designed and implemented an internal image similarity tool using perceptual hash (imagehash) and SSIM (scikit-image) to compare product cover images with availability images, supporting visual product matching and content validation.
- Automated image extraction and analysis pipeline using Selenium and Python, reducing manual matching workload by 80% and increasing recommendation accuracy by 25%.

SKILLS

Programming Languages: C++, CUDA, Python, C, Java, Dart, JavaScript, Unix Shell, SQL
Tools / Frameworks: AWS, Docker, Redis, NumPy, Pandas, Flask, Spring Boot, Vue.js, Flutter, Git, Sentry